

Question Number	Answer	Mark
6(a)(i)	The only correct answer is B heterozygosity index <i>A is incorrect because the Hardy-Weinberg equation is not used to calculate genetic diversity</i> <i>C is incorrect because index of diversity is not used to calculate genetic diversity</i> <i>D is incorrect because mitotic index is not used to calculate genetic diversity</i>	(1)

Question Number	Answer	Additional guidance	Mark
6(a)(ii)	An answer that makes reference to the following: genetic diversity considers one species whereas species richness considers {different / number of} species (1)	Mark as a whole Accept genetic diversity considers one population whereas species richness considers {one habitat / several populations}	(2)

	genetic diversity considers {alleles / genotypes} whereas species richness {is within a habitat / considers whole organisms / counts number of species (in an area)} (1)	Accept gene pool Ignore genes Accept specific area / ecosystem	
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Question Number	Answer	Additional guidance	Mark
6(b)	An answer that makes reference to two of the following: - can store more seeds as they {take up less space / are smaller} (1) - seeds need less {maintenance / cost} (1) - seeds can {survive longer than plants / be frozen } (1)	Accept growing plants takes up more space Accept more seeds stored aids maintenance of genetic diversity Accept converse for plants Accept seeds can be stored for longer	(2)

Question Number	Answer	Additional guidance	Mark
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6(c)(i)	A calculation showing the following steps: • correct difference (1) • correct percentage increase (1)	Mark the answer on answer line first <u>Example of calculation:</u> $8.2 - 3.3 / 8.2 \times 10^5 - 3.3 \times 10^5 / 4.9 / 4.9 \times 10^5 / 490000$ $(4.9 \div 3.3) \times 100 = 148 / 148.48 / 148.5$ Correct answer (148 / 148.48 / 148.5) with no working shown gains full marks Accept for one mark only 59.8 / 59.756 / 59.76 / 60	(2)
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Question Number	Answer	Additional guidance	Mark
6(c)(ii)	An explanation that makes reference to four of the following: • (all Cavendish plants) are susceptible to the fungus because they are {genetically identical / clones} (1) • whereas different varieties may have resistance to the fungus / contain an allele for resistance (to fungus) (1) • (therefore breeding could) increase genetic {diversity / variation} (1) • resulting in (new banana) plants with resistance to {fungus / Panama disease} (1)	Ignore mutations / immunity Accept do not contain an allele for resistance (to fungus) Accept contain an advantageous allele Ignore 'genes' unless qualified with alleles Accept increase gene pool Accept {offspring / new varieties} survive and reproduce	(4)